

```
import kagglehub

# Download latest version
path = kagglehub.dataset_download("kritikseth/fruit-and-vegetable-image-recognition")

print("Path to dataset files:", path)

↪ Path to dataset files: /kaggle/input/fruit-and-vegetable-image-recognition

import os

os.listdir(path)
folder_train = os.path.join(path, "train")
os.listdir(folder_train)
```

```
↪ ['capsicum',
  'sweetcorn',
  'orange',
  'tomato',
  'turnip',
  'ginger',
  'raddish',
  'pomegranate',
  'pineapple',
  'jalepeno',
  'apple',
  'carrot',
  'lettuce',
  'bell pepper',
  'eggplant',
  'beetroot',
  'kiwi',
  'pear',
  'cabbage',
  'cauliflower',
  'paprika',
  'lemon',
  'sweetpotato',
  'grapes',
  'cucumber',
  'corn',
  'banana',
  'garlic',
  'chilli pepper',
  'watermelon',
  'mango',
  'peas',
  'onion',
  'potato',
  'spinach',
  'soy beans']
```

```
import matplotlib.pyplot as plt

from PIL import Image
from collections import defaultdict

image_size = defaultdict(int)
train_folder = os.path.join(path, "train")
# img_path = os.path.join(train_folder, os.listdir(train_folder)[0])
# os.listdir(img_path)
for label in os.listdir(train_folder):
    # print(os.listdir(folder))
    for file in os.listdir(train_folder):
        list_file = os.listdir(os.path.join(train_folder, file))
        for img_path in list_file:
            finalPath = os.path.join(train_folder, file, img_path)
            img = Image.open(finalPath)
            # print(train_folder, file, img_pathr)

            # build_filepath = os.path.join(train_folder, file, img_pathr)
            try:
                image_size[img.size] += 1
```

```

except:
    pass
for size,count in image_size.items():
    print(f'{size} , {count}')
```

```

(600, 600) , 36
(768, 576) , 36
(256, 256) , 36
(387, 387) , 36
(280, 280) , 36
(1200, 822) , 36
(1920, 1280) , 36
(1400, 966) , 36
(1200, 628) , 72
(800, 600) , 36
(448, 375) , 36
(3008, 2000) , 36
(1500, 750) , 36
(2226, 2891) , 36
(600, 450) , 36
(2125, 1417) , 36
(500, 500) , 36
(1410, 2000) , 36
(1024, 768) , 36
(1200, 961) , 36
(800, 607) , 36
(1300, 1035) , 36
(1200, 1600) , 36
(2592, 1728) , 36
(800, 533) , 36
(1024, 415) , 36
(1582, 1186) , 36
(1200, 675) , 36
(1024, 1170) , 36
(1037, 692) , 36
(1050, 700) , 36
(720, 720) , 36
(1698, 1131) , 36
(640, 428) , 36
(900, 600) , 36
```

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Mã được tạo có thể phải tuân theo một giấy phép |

```

from tensorflow.keras.preprocessing.image import load_img, img_to_array, array_to_img
import tensorflow as tf
from google.colab import drive
```

```

myDrive = drive.mount('/content/drive')
save_path = '/content/drive/MyDrive/ml/GiuaKy/train_resized'
source_path = os.path.join(path, "train")
target_size = (256,256)

for label_folder in os.listdir(source_path):
    print(label_folder)
    pathlabel = os.listdir(os.path.join(source_path,label_folder))
    output_folder = os.path.join(save_path,label_folder)
    os.makedirs(output_folder, exist_ok=True)
    for img_file in pathlabel:
        pathImage = os.path.join(os.path.join(source_path,label_folder),img_file)
        print(pathImage)
        try:
            img = load_img(pathImage, target_size=target_size)
            print(f'test {img}')
            save_path_full = os.path.join(output_folder,img_file)
            img.save(save_path_full)
            print(f'save {save_path_full}')
        except Exception as e:
            print(f"Error with image {pathImage}: {e}")
```



```
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7976C0FF4610>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/mango/Image_22.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/mango/Image_53.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7976C0FF6690>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/mango/Image_53.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/mango/Image_69.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7976C0F50850>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/mango/Image_69.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/mango/Image_68.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7976C1010A90>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/mango/Image_68.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/mango/Image_88.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7976C1010F90>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/mango/Image_88.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/mango/Image_4.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7976C0FF4610>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/mango/Image_4.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/mango/Image_75.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7976C0F85ED0>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/mango/Image_75.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/mango/Image_12.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7976C0F50490>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/mango/Image_12.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/mango/Image_16.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7976C0F03550>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/mango/Image_16.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/mango/Image_62.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7976C4B36310>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/mango/Image_62.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/mango/Image_2.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7976C4CF6BD0>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/mango/Image_2.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/mango/Image_45.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7976C1A9EE10>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/mango/Image_45.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/mango/Image_7.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7976FBE4E690>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/mango/Image_7.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/mango/Image_85.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7976C0F84B90>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/mango/Image_85.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/mango/Image_5.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7976C1012210>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/mango/Image_5.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/mango/Image_86.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7976C0F67850>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/mango/Image_86.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/mango/Image_99.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7976C1ADFB50>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/mango/Image_99.jpg
```

```
print(source_path)
extensions = set()
for root, dirs, files in os.walk(source_path):
    for file in files:
        if '.' in file:
            extension = file.split('.')[1].lower()
            extensions.add(extension)

print(extensions)

/kaggle/input/fruit-and-vegetable-image-recognition/train
{'png', 'jpg', 'jpeg'}
```

Check size and total after resize

Number of images per class

```
labelfolder = os.listdir(save_path)

def count_images_in_folder(folder_path):
    counts = defaultdict(int)
    for label in os.listdir(folder_path):
        label_path = os.path.join(folder_path, label)
        if os.path.isdir(label_path):
            counts[label] = len(os.listdir(label_path))
    return counts
before = count_images_in_folder(source_path)
print(before)
```

```
after = count_images_in_folder(save_path)
print(after)
```

```
defaultdict(<class 'int'>, {'capsicum': 89, 'sweetcorn': 91, 'orange': 69, 'tomato': 92, 'turnip': 98, 'ginger': 68, 'raddish': 81, 'porr
defaultdict(<class 'int'>, {'capsicum': 89, 'sweetcorn': 91, 'orange': 69, 'tomato': 92, 'turnip': 98, 'ginger': 68, 'raddish': 81, 'porr
```

Compare image dimensions

```
def get_image_size(folder_path):
    sizes = defaultdict(int)
    for label in os.listdir(folder_path):
        label_path = os.path.join(folder_path, label)
        if os.path.isdir(label_path):
            for img_file in os.listdir(label_path):
                img_path = os.path.join(label_path, img_file)
                img = Image.open(img_path)
                sizes[img.size] += 1
    return sizes
print(source_path)
print(save_path)
before = get_image_size(source_path)
print(sum(before.values()))

after = get_image_size(save_path)
print(sum(after.values()))

/kaggle/input/fruit-and-vegetable-image-recognition/train
/content/drive/MyDrive/ml/GiuaKy/train_resized
3115
2488
```

Bắt đầu lập trình hoặc [tạo](#) mã bằng trí tuệ nhân tạo (AI).